# Experiment No -: 07

Aim : To study the controlability and observability of Given system.

Given system :

$a = [-1.5, -2, 1, 0];$

$b = [0.5, 0]$

$c = [0, 1]$

$d = 1;$

Solution :

$A = \begin{bmatrix} -1.5 & -2 \\ 1 & 0 \end{bmatrix}$ ; $B = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$ ; $C = [0 \ 1]$ ; $D = [1]$

To check controlability

A is 2X2 Matrix

$\therefore n = 2$

$QC = [B : AB]$

$B = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$ $AB = \begin{bmatrix} -1.5 & -2 \\ 1 & 0 \end{bmatrix}$

$|QC| = \begin{bmatrix} 0.5 & -0.75 \\ 0 & 0.5 \end{bmatrix}$

$|QC| = 0.25$

Rank of $QC = 2 = n$

$\therefore$ hence System is controllable.

$$Q_o = [C^T \; : \; A^T C^T] \quad \text{for} \quad n = 2$$

$$\therefore C^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad A^T = \begin{bmatrix} -1.5 & 1 \\ -2 & 0 \end{bmatrix}$$

$$C^T A^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} -1.5 & 1 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Q_o = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|Q_o| = -1 \neq 0$$

Hence the system is observable.

# Experiment No: 07

Aim : To ~~study~~ find Controllability and observability of control system.

CODE -

```
a = [-1.5,-2;1,0];
b = [0.5;0];
c = [0,1];
d = 1;
sys = ss(a,b,c,d);

co = ctrb(sys);
rank(co)

ob = obsv(sys);
rank(ob)
```

OUTPUT-

Command Window

```
>> exp_7

ans =

     2


ans =

     2

fx >>
```

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# Experiment No. 08

**Aim:** To design a lead compensator using root locus and achieve desired specification.

**Code:**

```
clc;
clear;
close all;
s=tf('s');
Gs=1/s/(s+1);
figure(1)
step(Gs);
title("unit step response of the open loop system");
Mp=0.04;
ts=3.5;
z=(log(Mp)/pi)^2;
xi=sqrt(z/(1+z));
wn=4/(xi*ts);
figure(2)
sd= -xi*wn+wn*sqrt(1-xi^2)*1i;
plot(real(sd),imag(sd),'mx','Linewidth',2);
hold on;
rlocus(Gs);
[num,den]=tfdata(Gs, 'v');
n=length(den);
denSd=0;
for k=1:n
denSd=denSd+den(k)*sd^(n-k);
end
numSd=0;
for k=1:n
numSd=numSd+num(k)*sd^(n-k);
end
phi=0-(pi+atan(imag(denSd)/real(denSd)));
phi_deg=phi*180/pi; phic=-pi-phi;
phic_deg=1-80-phi_deg
theta=acos(xi); %rad
gamma=0.5*(pi-theta-phic); %rad
zc=wn*sin(gamma)/sin(pi-theta-gamma);
```

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```
pc=wn*sin(gamma+phic)/sin(pi-theta-gamma-phic);
alpha=zc/pc;
gc=(s+zc)/(s+pc);
figure(3)
sd=-xi*wn+wn*sqrt(1-xi^2)*1i;
plot(real(sd),imag(sd), 'mx', 'LineWidth',2);
hold on;
rlocus(Gs*gc)
Kc=7.1;
[numc1,dencl]=tfdata(gc*Gs,'v');
n=length(den);
denSdc1=0;
for k=1:n
denSdc1=denSdc1+den(k)*sd^(n-k)
end
numSdc1=0;
for k=1:n
numSdc1=numSdc1+num(k)*sd^(n-k);
end
Kc=abs(denSdc1/numSdc1);
Gc=Kc*gc;
Gcl= minreal((Gs*Gc)/(1+Gs*Gc));
figure(4)
step(Gcl)
```

**Output:**

```
phic_deg =
  153.9946

denSdc1 =
   0.0620 - 2.5495i

denSdc1 =
  -1.0809 - 1.4341i

denSdc1 =
  -1.0809 - 1.4341i
```

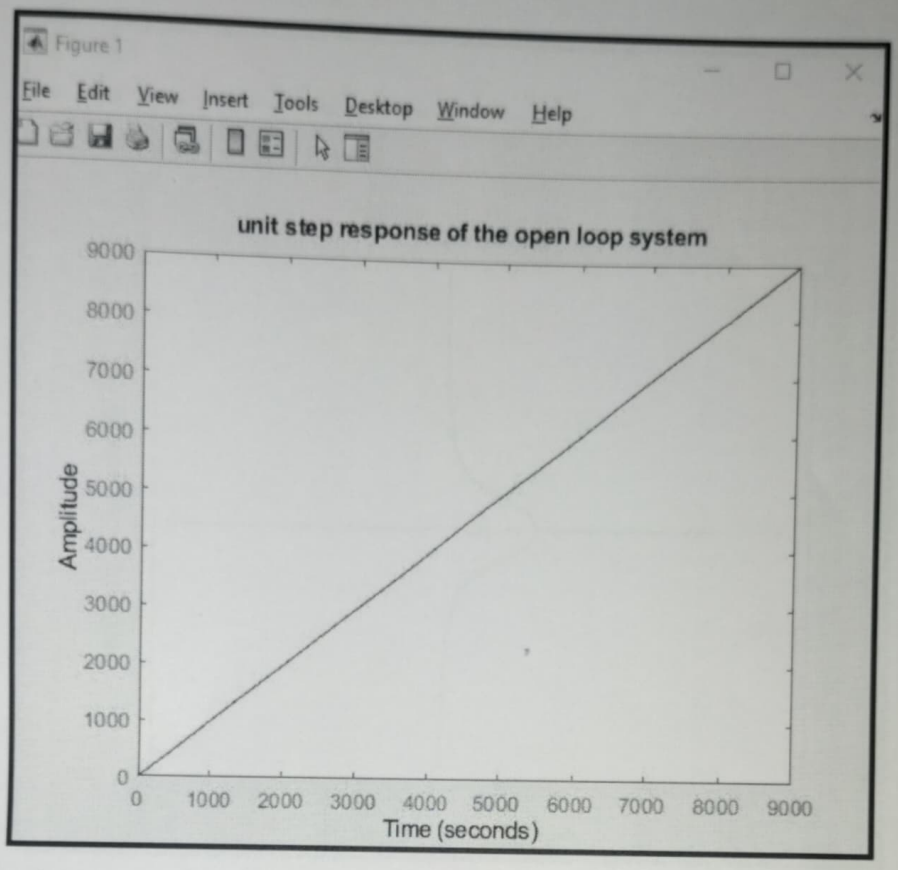
Figure 1
File Edit View Insert Tools Desktop Window Help
unit step response of the open loop system
Amplitude
Time (seconds)
0
1000
2000
3000
4000
5000
6000
7000
8000
9000

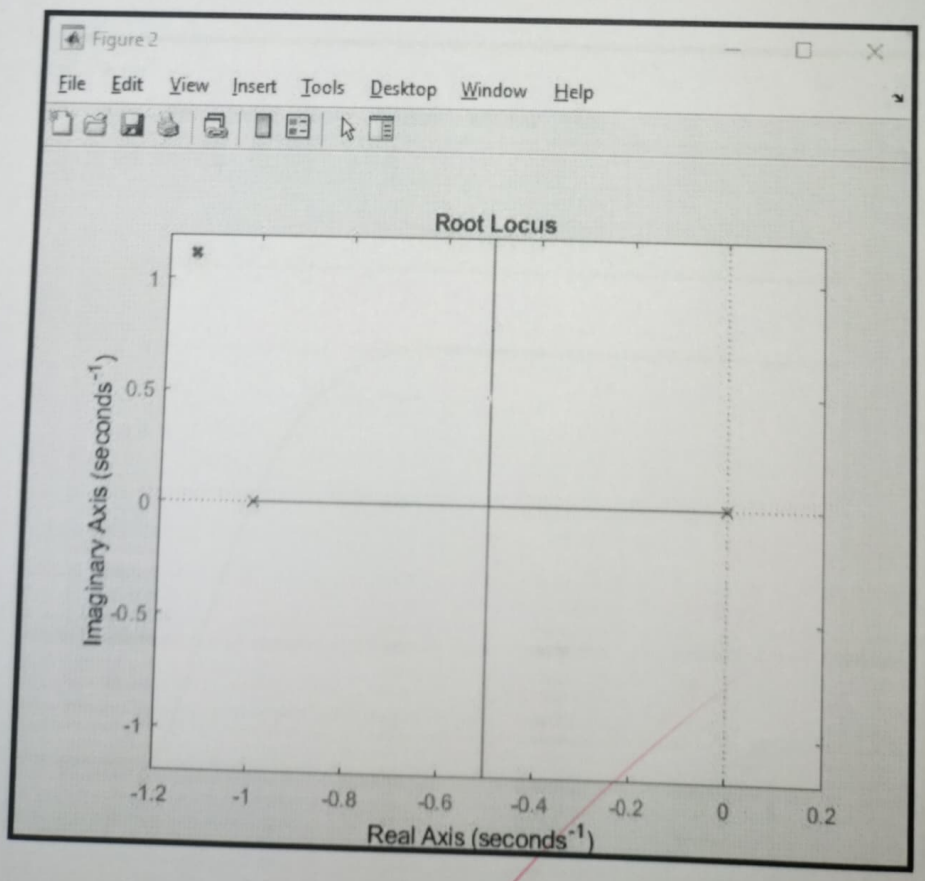
Figure 2
File Edit View Insert Tools Desktop Window Help
Root Locus
Imaginary Axis (seconds$^{-1}$)
Real Axis (seconds$^{-1}$)
1
0.5
0
-0.5
-1
-1.2
-1
-0.8
-0.6
-0.4
-0.2
0
0.2

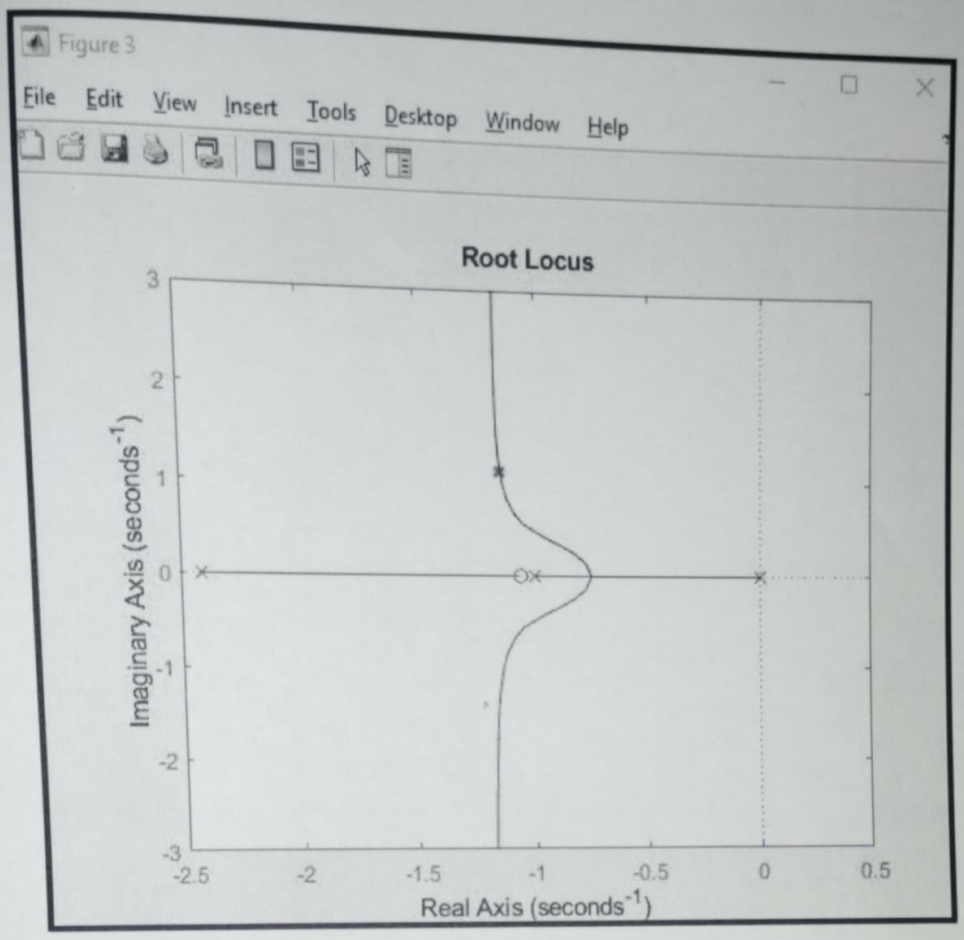
Figure 3
File Edit View Insert Tools Desktop Window Help
Root Locus
Imaginary Axis (seconds$^{-1}$)
Real Axis (seconds$^{-1}$)
3
2
1
0
-1
-2
-3
-2.5
-2
-1.5
-1
-0.5
0
0.5

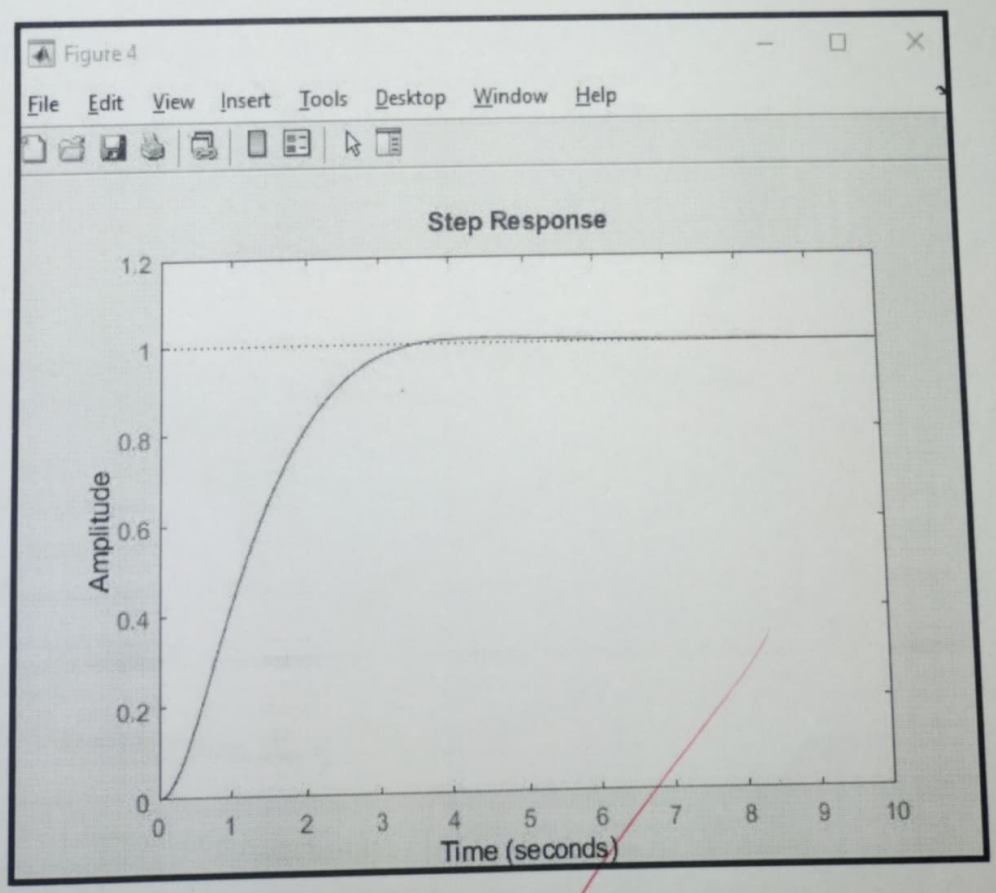
Figure 4
File Edit View Insert Tools Desktop Window Help
Step Response
Amplitude
Time (seconds)
1.2
1
0.8
0.6
0.4
0.2
0
0
1
2
3
4
5
6
7
8
9
10

# Experiment No.09

**Aim:** To design the state feedback matrix for the given system to achieve the desired response.

**Code:**

```
clc; clear; close all;
% % System - 1
A =[0 1 0; 0 0 1;1 -3 2];
B =[0 0 2]';
% Step 01: Check the controllable
Qc = ctrb(A, B);
if rank(Qc)==length(A)
   disp('The system-1 is full state controllable')
else
   disp('The system-1 is not full state controllable')
end
%Step 02: Obtain the closed-loop characteristic polynomial
syms s k1 k2 k3; % declare the symbols
% state feedback matrix
K = [k1 k2 k3];
%closed loop matrix
Ac =A-B*K
%closed-loop characteristic polynomial;
Pd=charpoly(Ac)
%Desired pole location -1,-2,-3
Pc = poly([-1 -2 -3])
%Step 03: Obtain state feedback matrix by comparing desired and
% closed-loop characteristic polynomial
delta=Pc-Pd; %LHS-RHS=0
sol=solve(delta, [k1 k2 k3])
k1 = double(sol.k1);
k2 = double(sol.k2);
k3 = double(sol.k3);
%state feedback matrix
K = [k1 k2 k3]
disp('closed loop eigenvalues are: ')
eig(A-B*K)
% % Open and Closed Loop responsed;
% assume y=x1; So
```

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```
C = [1 0 0]; D = 0;
% Open loop System
sys=ss(A, B, C, D)
%Step Response of Open loop system
figure(1)
step(sys);
%closed loop system
sysc=ss(A-B*K,B,C,D)
%step response
figure(2)
step(sys)
% % Alternate method
% Step 02: Compute state feedback matrix
K1 = place(A, B,[-1 -2 -3]);
eig(A-B*K
```

**Output:**

```
The system-1 is full state controllable

Ac =

[     0,     1,     0]
[     0,     0,     1]
[1 - 2*k1, - 2*k2 - 3, 2 - 2*k3]


Pd =

[1, 2*k3 - 2, 2*k2 + 3, 2*k1 - 1]


Pc =

   1   6  11   6


sol =

  struct with fields:
```

```
k1: 7/2
k2: 4
k3: 4


K =

   3.5000    4.0000    4.0000

closed loop eigenvalues are:

ans =

  -1.0000
  -2.0000
  -3.0000


sys =

  A =
       x1  x2  x3
   x1   0   1   0
   x2   0   0   1
   x3   1  -3   2

  B =
       u1
   x1   0
   x2   0
   x3   2

  C =
       x1  x2  x3
   y1   1   0   0

  D =
       u1
   y1   0
```

Continuous-time state-space model.

```
sysc =

  A =
       x1  x2  x3
   x1   0   1   0
   x2   0   0   1
   x3  -6 -11  -6

  B =
       u1
   x1   0
   x2   0
   x3   2

  C =
       x1  x2  x3
   y1   1   0   0

  D =
       u1
   y1   0
```

Continuous-time state-space model.

```
ans =

   -1.0000
   -2.0000
   -3.0000

>>
```

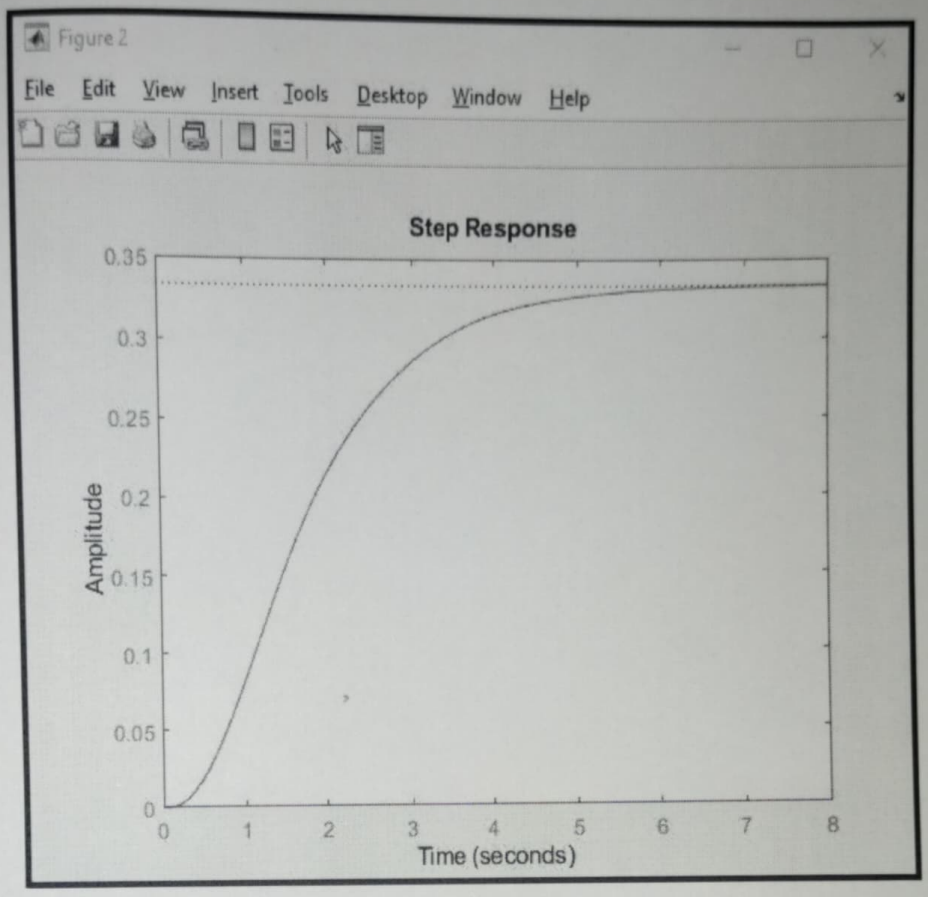


Figure 2
File Edit View Insert Tools Desktop Window Help
Step Response
Amplitude
Time (seconds)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0 1 2 3 4 5 6 7 8

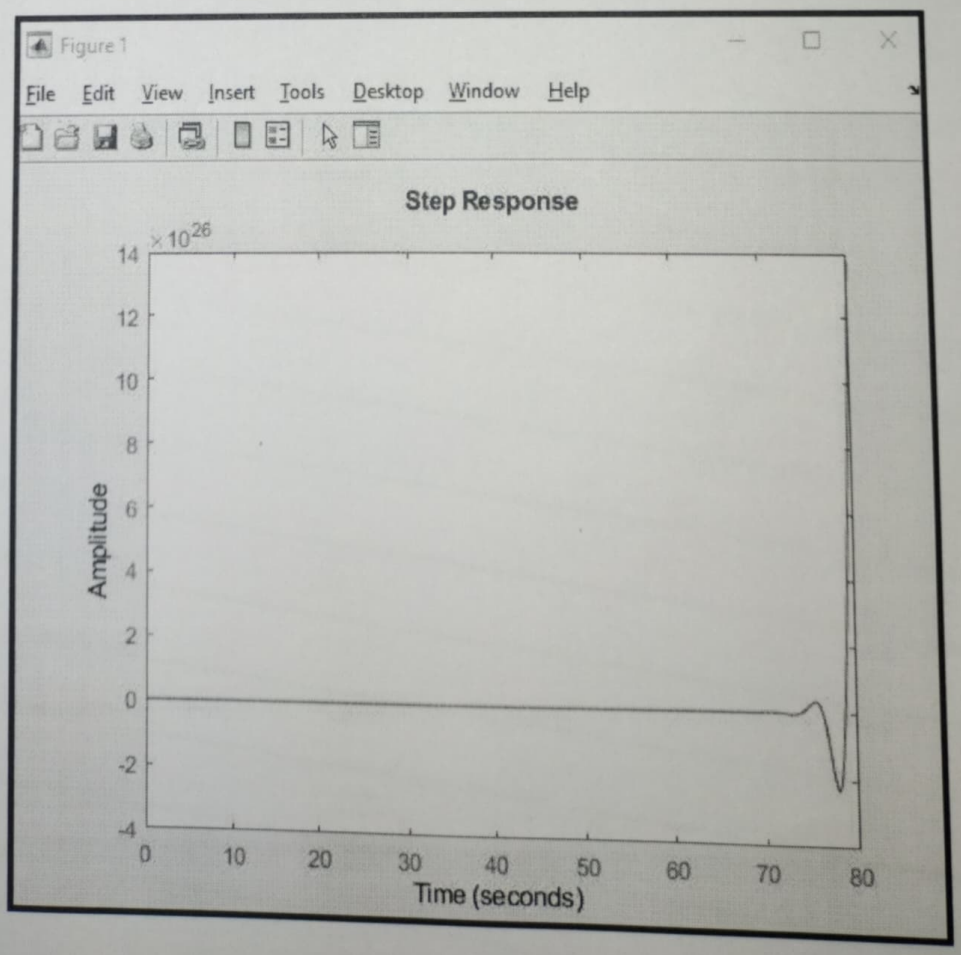


Figure 1
File Edit View Insert Tools Desktop Window Help
Step Response
×10^26
Amplitude
Time (seconds)
14
12
10
8
6
4
2
0
-2
-4
0 10 20 30 40 50 60 70 80

EXPERIMENT NO : 10

Name : Soham Malgundkar

Roll no : 54

sub : PCS

* Find Nyquist Plot from Given Transfer function

$$G(S)\ H(S) = \frac{1}{S(S+2)}$$

→ Position of Poles $S=0$, $S=-2$

No Poles on RHP, $P=0$

∴ Encirclement about $(-1+0j)$ Point in C.C.W direction

Now Pole is at origin

Nyquist Paths are

Path a-b, $S = j\omega$

Path b-c-d ; $S = \lim_{R\to\infty} Re^{j\theta}$

Path d-e ; $S = -j\omega$

Path e-f-a ; $S = \lim_{\epsilon\to 0} \epsilon e^{j\theta}$

Path a-b

$$G(S)\,H(S) = \frac{1}{S(S+2)}$$

$$G(j\omega)\,H(j\omega) = \frac{1}{j\omega(j\omega+2)}$$

$$|G(j\omega)\,H(j\omega)| = \frac{1}{j\omega(j\omega+2)}$$

$$|G(j\omega)\,H(j\omega) = \frac{1}{\omega\sqrt{\omega^2+2}}$$

$$\angle G(j\omega)\,H(j\omega) = \tan^{-1}(0/1)$$

$$\tan^{-1}(\omega/0) + \tan^{-1}(\omega/2)$$

$$\angle G(j\omega)\,H(j\omega) = -90^\circ - \tan^{-1}(\omega/2)$$

At $\omega = 0$ $\quad |G(j\omega)\,H(j\omega)| = \infty$

$\angle G(j\omega)\,H(j\omega) = -90^\circ$

At $\omega = \infty$ $\quad |G(j\omega)H(j\omega)| = 0$

$\angle G(j\omega)H(j\omega) = -180^\circ$

Path b-c-d

$$s = \lim_{R \to \infty} R e^{j\theta}$$

$$G(s)H(s) = \frac{1}{Re^{j\theta}(Re^{j\theta}+1)}$$

This will be 0 is at $R \to \infty$

Path b-c-d maps ~~tan~~ on origin.

Path d-e

$s = -j\omega$

This is mirror image of section a-b

Path e-f-a

$$s = \lim_{\epsilon \to 0} \epsilon e^{j\theta}$$

$$G(s)H(s) = \frac{1}{\epsilon e^{j\theta}(\epsilon e^{j\theta}+1)}$$

$\epsilon \to 0$

$$G(s)H(s) = \frac{1}{\epsilon e^{j\theta}}$$

$$= \infty e^{-j\theta}$$

At Point $e$, $\theta = -90^\circ$ $\quad \therefore \infty e^{90j}$

$f$, $\theta = 0^\circ$ $\quad \therefore \infty e^{0j}$

$a$, $\theta = 90^\circ$ $\quad \therefore \infty e^{-90^\circ}$

Now $N = 0$ $\quad \therefore N = P - Z$

$Z = 0$ $(\because P = 0)$

Hence System is stable.

**Aim:** **To find the Nyquist Plot from given Transfer Function.**

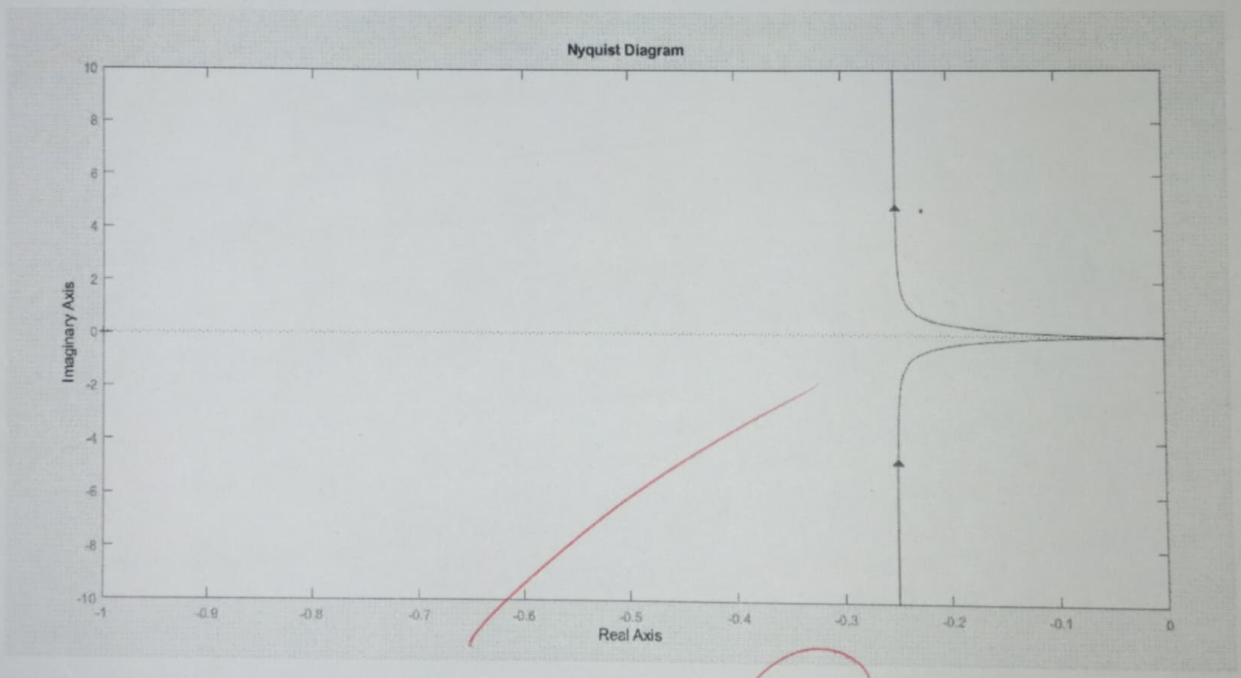
**Given T.F. :** **G(s).H(s) = 1 / S(S+2)**

**Code:**

```
%%Nyquist plot = G(s)H(s) = 1 / s^2+2s %%

sys = tf(1,[1,2,0]);
n = nyquistplot(sys);
```

**Output:**

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